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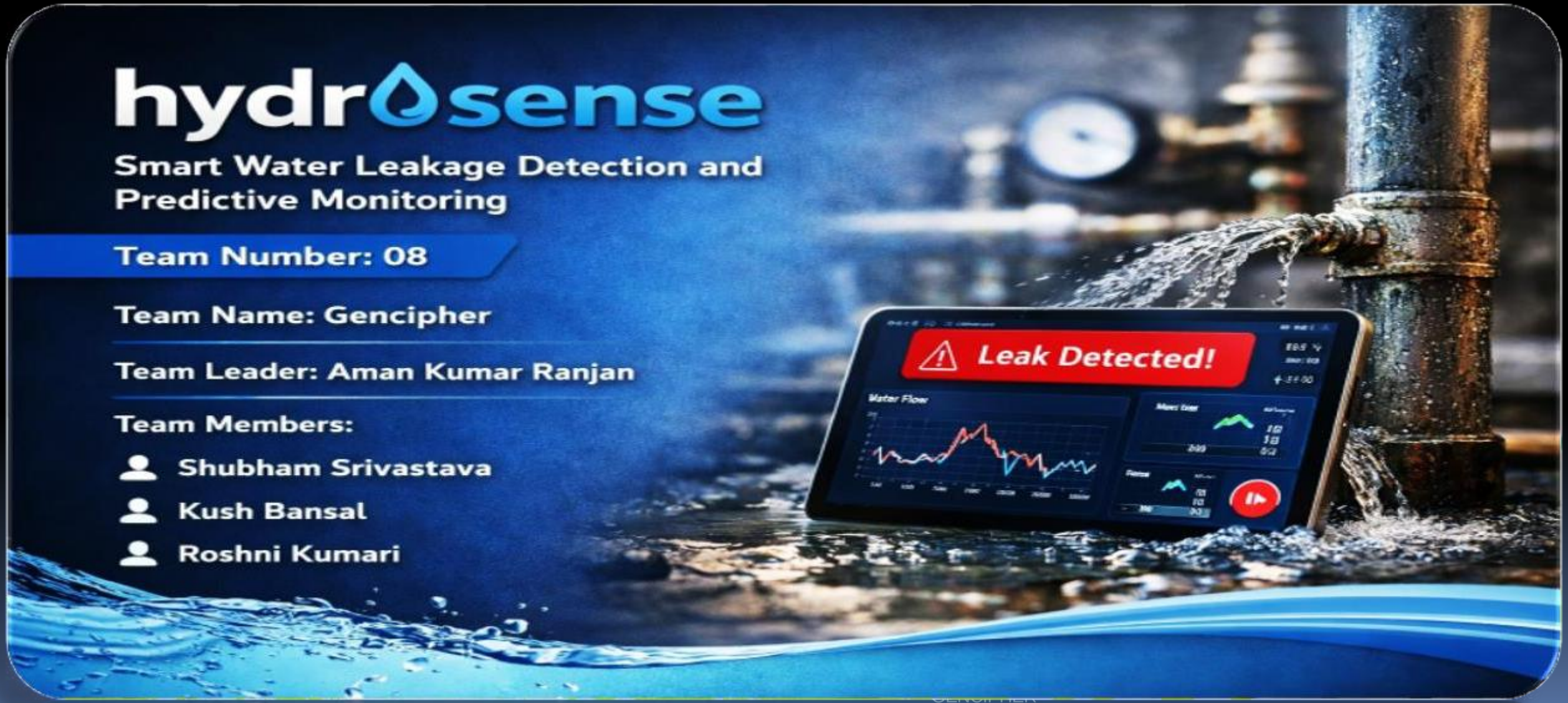
TECH SANGRAM



PROBLEM STATEMENT

- Problem Statement:

➢ Urban water systems lose 20–40% water due to leaks, theft, and delayed detection, with no real-time monitoring or precise fault localization—requiring an AI-powered IOT solution to detect, locate, and predict losses efficiently.

- Description:

➢ Current systems are costly, depend on manual inspection, and lack accurate leak detection, real-time monitoring, and scalability. They also face issues like poor localization and underutilized data. Despite initiatives like Jal Jeevan Mission, AMRUT Scheme, and Smart Cities Mission, implementation remains limited, highlighting the need for a low-cost, AI-powered IoT solution .

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PROPOSED SOLUTION

- **A smart IOT-based system using ESP8266 monitors water flow and detects leakage in real-time using intelligent data analysis**
- **Core Idea:**

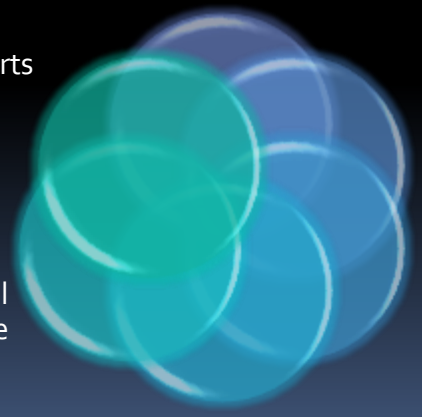
Instead of reacting after a leak occurs, the system continuously observes water behavior patterns and identifies subtle anomalies that indicate potential

Uses flow sensor (and optional pressure sensor) to continuously monitor water usage

ESP8266 (NodeMCU) collects sensor data and sends it to the cloud via Wi-Fi

Cloud platform stores and processes data for analysis

- **Working Mechanism**



Triggers instant alerts (LED / buzzer)

Sensors measure water flow and pressure

Detects abnormal patterns (possible leak or misuse)

ESP8266 sends real-time data to the cloud

System analyzes data using rule-based logic / AI model

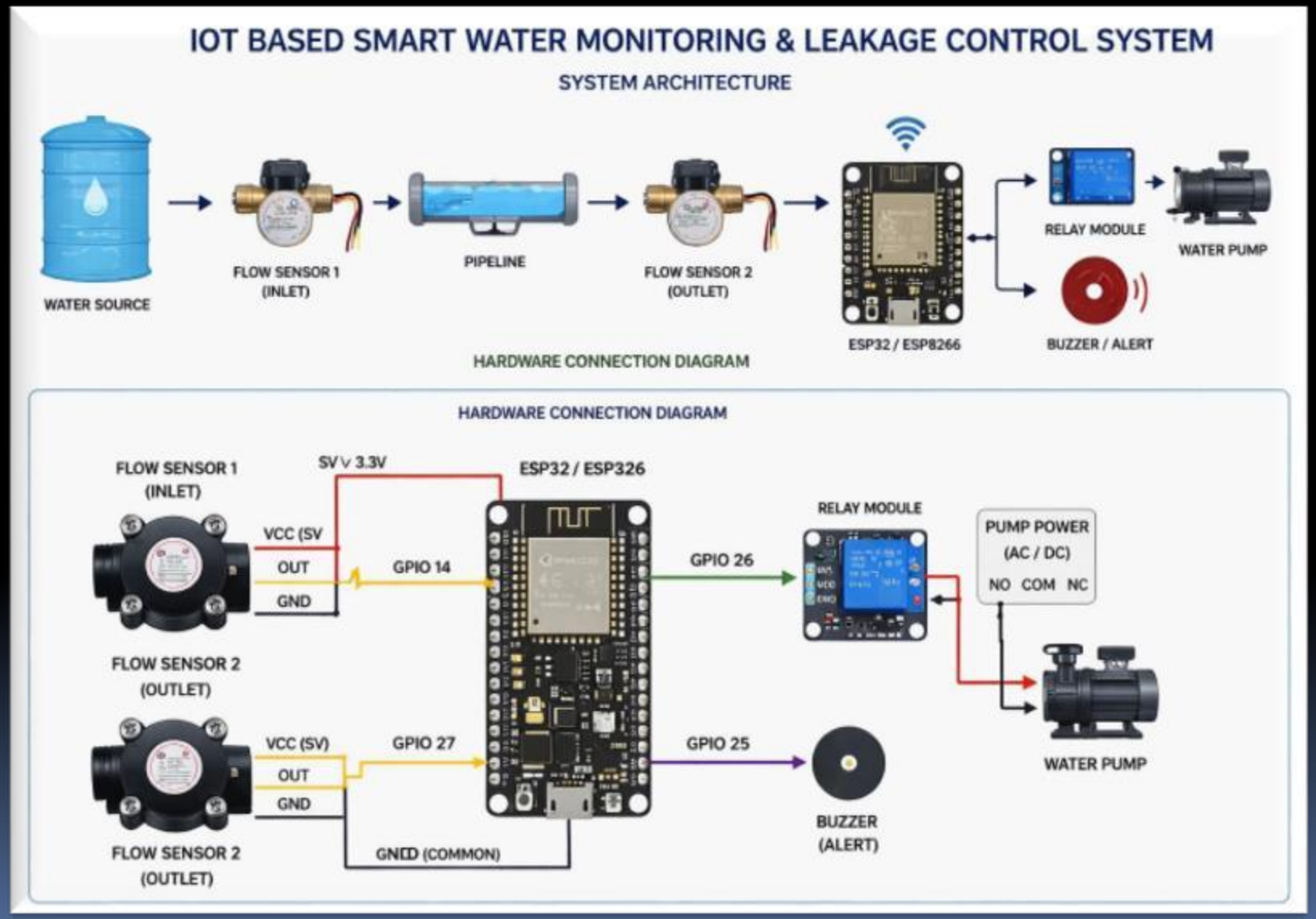
GENCIPHER

SYSTEM ARCHITECTURE

Real-time water monitoring system using dual flow sensors and ESP. Detects leakage by comparing inlet and outlet flow, automatically stops pump via relay, and triggers alerts. Supports IOT cloud integration for remote monitoring

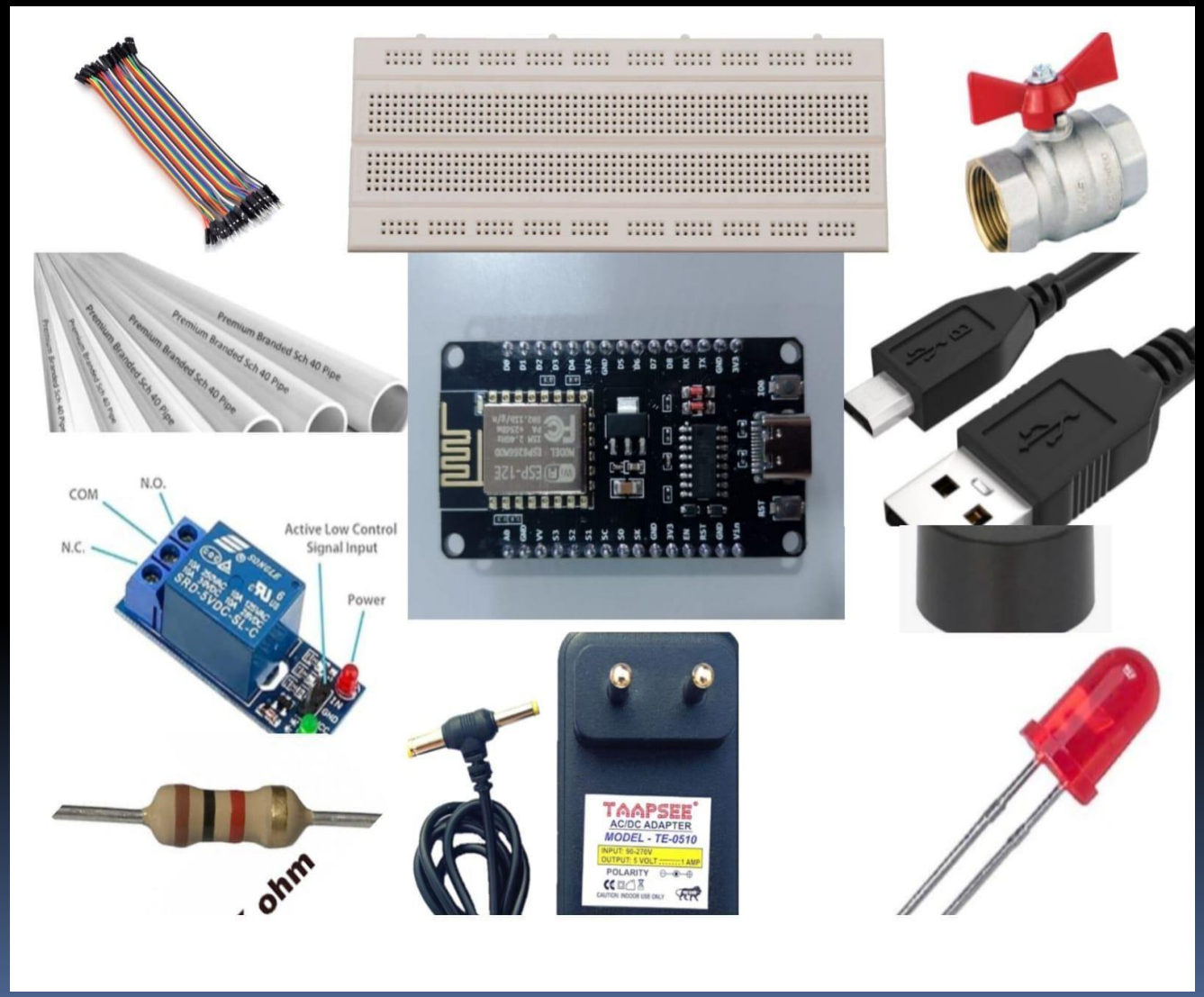
Hardware Used:

- ESP8266 / ESP32
- Flow Sensors ×2
- Relay Module
- Water Pump
- Buzzer
- Power Supply
- Jumper Wires / Breadboard



Key Components :

1. **ESP8266 :** Acts as the controller and Wi-Fi module to process sensor data and enable real-time monitoring and alerts.
2. **Flow Sensors :** Measure water flow at different points to detect discrepancies, helping identify leaks or theft.
3. **Relay Module:** Automatically controls the water pump (ON/OFF) based on detected anomalies or system logic.
4. **Buzzer:** Provides instant alerts when leakage or abnormal flow is detected.
5. **Power Supply :**Ensures stable operation of all components for continuous monitoring.
6. **Jumper Wires / Breadboard:** Enable easy circuit connections and prototyping without permanent soldering.



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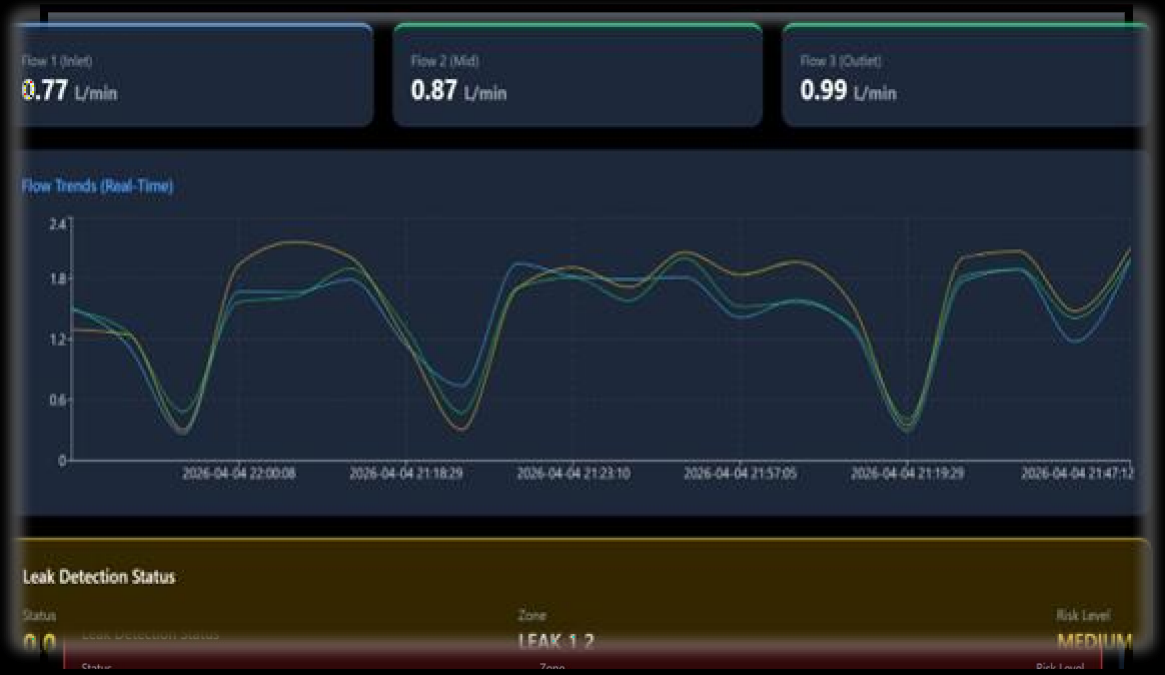
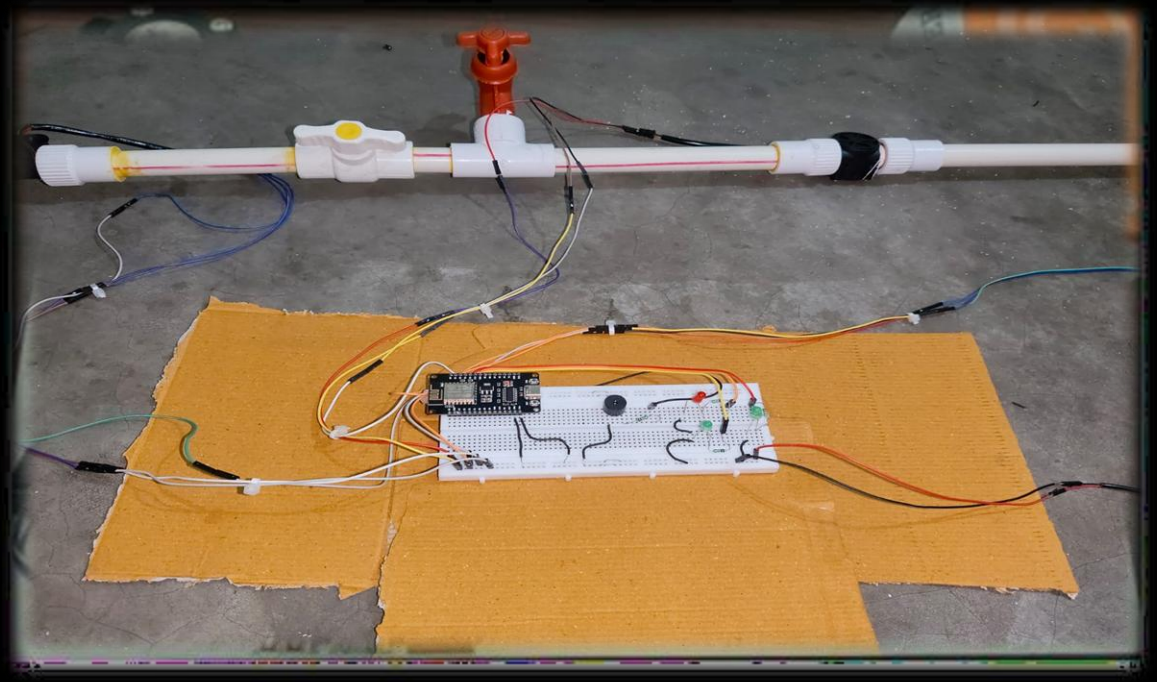
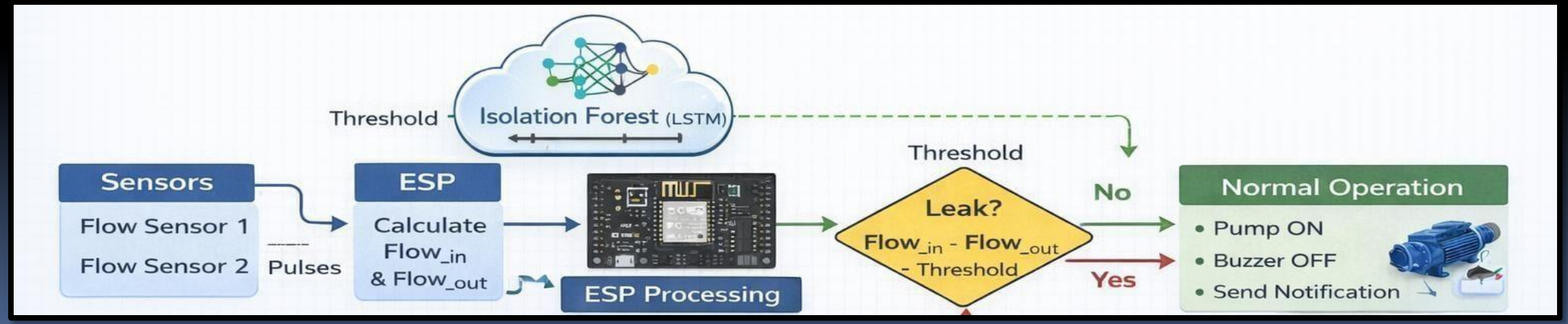


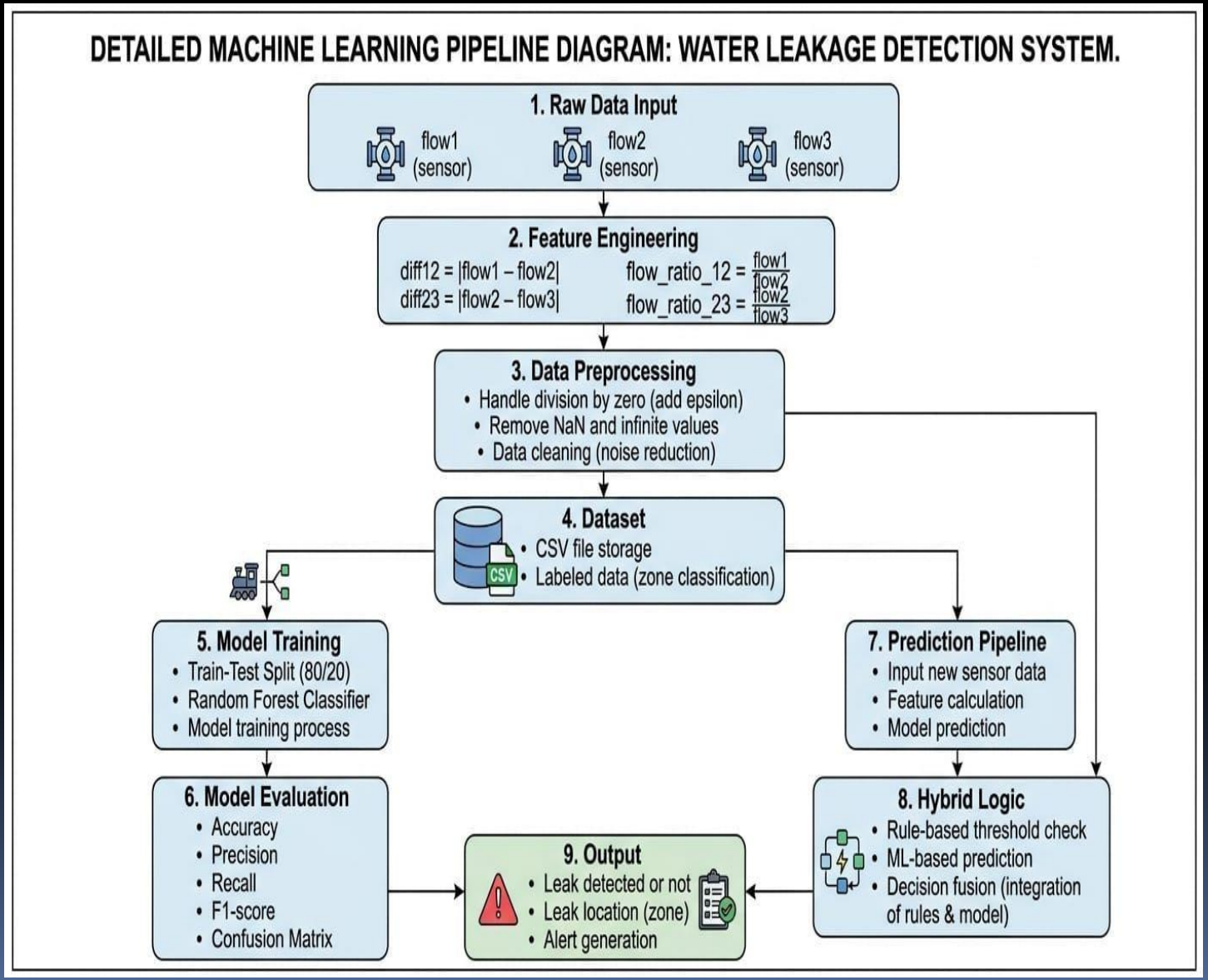




GENCIPHER

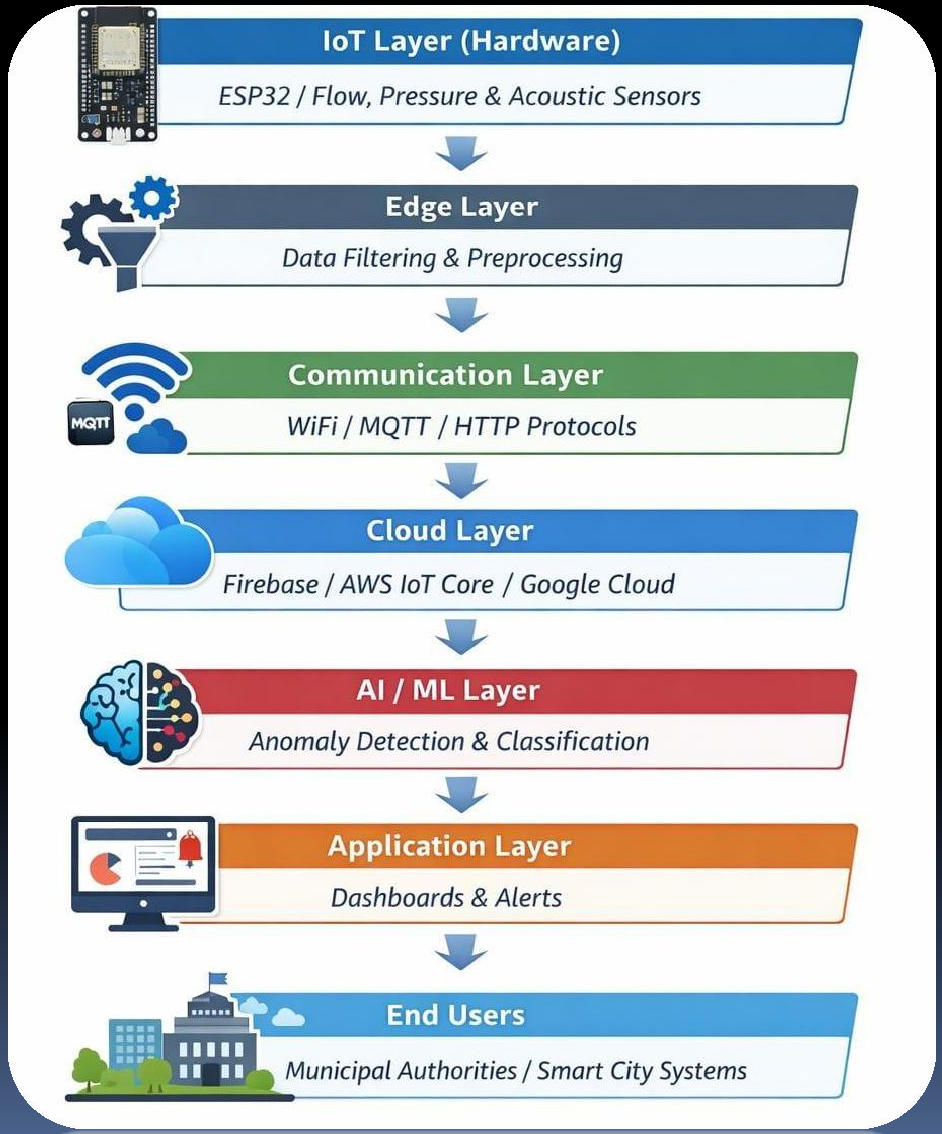
TECHNICALAPPROACH

- Collect real-time sensor data(flow1, flow2, flow3)
- Perform feature engineering(diff12, diff23, flow ratios)
- Preprocess data (handle NAN , infinity, division errors)
- Store cleaned data as labeled dataset (CSV)
- Train Random Forest model(80:20 train-test split)
- Evaluate using accuracy , precision, recall, F1-score
- Apply same pipeline to new incoming data
- Use hybrid logic (rule-based+ ML prediction)
- Output leak status, location , and trigger alerts



TECHNOLOGY STACK

- IoT layer uses ESP8266 sensors to collect real-time water data.
- Edge layer performs data filtering & preprocessing.
- Communication via WFI / MQTT / HTTP protocols.
- Cloud layer handles data storage & processing.
- AI/ML models perform leak & theft detection
- Application layer provides dashboard & real-time alerts
- End users (authorities) monitor and take quick action



Hydro Sense VS Existing Solutions

Existing Systems:

- High dependency on **dense sensor networks** → expensive infrastructure
- **Manual inspection & delayed detection** → slow response time
- **Poor leak localization** → difficult to pinpoint exact fault location
- High **false positives / false alarms**
- Limited scalability for **smart city deployments**
- Requires **large labeled datasets** for accurate performance
- **Difficult maintenance** (especially underground pipelines)
- Minimal integration with initiatives like **Jal Jeevan Mission, AMRUT Mission, Smart Cities Mission**

Hydro Sense

- **Low-cost architecture** using minimal sensors (optimized placement)
- **Real-time detection** using AI + IOT (no manual dependency)
- **Accurate leak localization** using multi-sensor comparison with (Zone Accuracy~97% and Severity Accuracy~96%)
- Reduced false alarms via **intelligent anomaly detection (LSTM / Isolation Forest)**
- Scalable for **city-wide deployment**
- **Edge AI (ESP32)** → faster response, reduced cloud load
- Seamless integration with **government dashboards & smart infrastructure**
- Supports **theft detection + predictive analytics** (not just detection)

Impact and Benefits

Environmental Impact

- Reduces **20–40% water loss** in urban pipelines
- Promotes **sustainable water management**
- Helps conserve **groundwater & natural resources**

Economic Benefits

- Minimizes **revenue** loss due to leakage & theft

- Reduces **maintenance & operational** costs
- Enables cost-**efficient** infrastructure (fewer sensors, edge AI)

Operational Efficiency

- **Real-time monitoring & alerts** ⟶ faster response
- Accurate **leak localization** ⟶ reduced repair time
- Lower **false alarms** with **intelligent AI** models

Smart City Enablement

- Scalable for **large urban deployments**

- Integrates with initiatives like **Smart Cities Mission, Jal Jeevan Mission**
- Supports **data-driven governance & planning**

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Research and References

- Research Work:

- International journal of Computational Learning and Intelligence:Smart Water Leakage Detection And Prevention System Using IOT Technology
- International Journal of Research Publication and Reviews: IOT Based Water Pipeline Leakage Detection System by
- International journal of progressive research in engineering management and science (ijprems):review paper on IOT based water pipeline leakage detection system